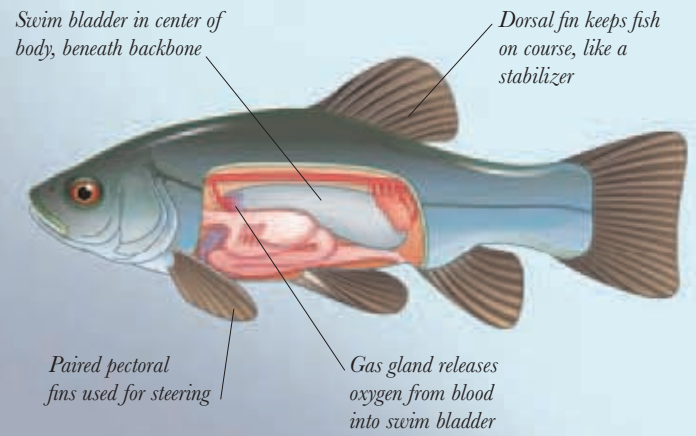


Staying up

For fish, staying at the right depth is important for survival. To control this, rays and sharks use their fins, and sharks have an oil-filled liver, which helps to buoy them up. But most fish use a swim bladder to control their buoyancy. The fish can rise or sink by altering the amount of gas in the swim bladder. It takes oxygen from its blood and releases it from a gas gland, blowing the swim bladder up like a balloon.



Fast fish

With their tapering bodies and scissor-shaped tails, these fusilier fish are built for speed. When they swim, their bodies stay stiff and nearly all the movement comes from their rapidly flicking tails. This way of swimming is also used by some much larger fish that reach record speeds. Over short distances, the fastest fish is the sailfish, which can swim at more than 60 mph (100 kph).

Pectoral fin helps generate lift

Wave peak reaches center of body length

Tail flicks left as wave moves along

Head swings right, starting S-shaped wave

Supple swimmers

Most sharks—including this dogfish—move by bending their bodies into curves. As the fish bends, the rear half of its body pushes against the water, with its tail fin adding a final flick. A shark's pectoral fins are slanted upward, which generates lift as it swims. Open-water sharks usually spend all their lives swimming. This keeps water flowing over their gills and helps their blood to circulate.